

EDUCATION

Lovely Professional University
Bachelor of Technology (Computer Science & Engineering)

Jalandhar, India
July 2022 - May 2026

SKILLS SUMMARY

- **Programming:** Python, SQL
- **Machine Learning:** Machine Learning, Deep Learning, Statistics & Probability, Model Training, Model Evaluation, Supervised Learning, Unsupervised Learning
- **AI Domains:** Natural Language Processing, Computer Vision, Medical Imaging, Generative AI, Large Language Models
- **ML Frameworks:** PyTorch, Scikit-learn, Hugging Face
- **Data:** Data Preprocessing, Document Processing, Data Collection, Data Labeling, Feature Engineering

WORK EXPERIENCE

- AI Intern | Know AI | [LINK](#)

June 2026 - August 2026

 - Built Generative AI and LLM applications using Python, focusing on prompt tuning and evaluation.
 - Developed and tested AI workflows and REST APIs using LLMs, APIs, tool calling, automation, and RAG.
 - Designed **RAG pipelines** using embeddings, vector databases, and semantic search for knowledge-grounded retrieval.
 - Developed LLM and agent evaluation pipelines measuring accuracy, reliability, tool performance, and output quality.
 - Improved AI reliability through monitoring, troubleshooting, debugging, latency optimization, and failure handling.
- Data Engineer Intern | flickDONE | [LINK](#)

June 2025 - August 2025

 - Built scalable Python data pipelines processing 50,000+ records for preprocessing and feature engineering.
 - Engineered automated **data preprocessing and validation workflows**, reducing manual verification by 70%.
 - Optimized AWS-based AI/ML pipelines, improving execution speed by 25% through performance tuning.
 - Developed reusable AI workflows for document processing, LLM applications, API integrations, and enterprise automation.
 - Collaborated with teams to build scalable AI/ML workflows and improve system reliability and deployment.

PROJECTS

- LangGraph Workflow Orchestrator — Agentic AI Platform | [LIVE](#)

June 2026 – August 2026

 - Built and deployed a stateful **Agentic AI platform** using Python, LangGraph, LangChain, FastAPI, and Gemini.
 - Designed agentic workflows for task routing, tool execution, automation, failure handling, and recovery.
 - Developed modular components for **task decomposition, tool calling, function calling, and agent coordination**.
 - Exposed AI workflows through FastAPI REST APIs, enabling scalable deployment and integration with external services.
 - Evaluated agent workflows using evaluation datasets, multi-turn scenarios, failure analysis, and performance benchmarking.
- Production-Grade RAG Engine | [LIVE](#)

June 2026 - August 2026

 - Built an end-to-end production-grade RAG system using Python, FastAPI, LangChain, Qdrant, and AWS.
 - Developed document preprocessing, chunking, embeddings, indexing, retrieval, and response generation pipelines.
 - Implemented **semantic vector search with Qdrant** for low-latency, knowledge-grounded context retrieval.
 - Added reranking and retrieval evaluation to improve retrieval quality, context relevance, and AI response performance.
 - Improved AI service reliability through testing, monitoring, fallback handling, and error analysis.
- Geospatial Deep Learning Segmentation | [LINK](#)

January 2026 - March 2026

 - Built an end-to-end deep learning segmentation pipeline using Python, PyTorch, U-Net, and computer vision techniques.
 - **Developed image preprocessing and annotation workflows** for satellite imagery datasets and segmentation masks.
 - Trained and evaluated deep learning models for pixel-level building and road segmentation from geospatial imagery.
 - Built inference pipelines and API workflows for processing images and generating reliable segmentation prediction outputs.
 - Generated **GeoJSON and GeoTIFF outputs** for infrastructure mapping and geospatial analysis.

CERTIFICATES

- Python, Data Science & ML (Cipherschools) | [CERTIFICATE](#)

August 2024

 - Built machine learning models using PyTorch and applied ML techniques for model development and evaluation.
 - Developed **Dockerized environments** supporting reproducible experimentation and deployment
- Generative AI with Large Language Models (Coursera) | [CERTIFICATE](#)

March 2024

 - Applied prompt engineering techniques for production LLM applications.
 - Built **retrieval workflows, AI output evaluation, and RAG pipeline** performance validation.